Yulong Ge

Ph.D. Student in Applied Mathematics | AI for Science

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CV Downloads

- English PDF | 中文 PDF

Education

Ph.D. **Harbin Institute of Technology**, Applied Mathematics — Harbin, China
- Ph.D. student at the School of Mathematics.
- Supervised by Professor Jianwei Ma.
- Research focuses on computer vision, generative models, neural network interpretability, and AI for Science.

Zhongguancun Academy, Doctoral Joint Training — Beijing, China

B.S. **Harbin Institute of Technology**, Information and Computing Science — Harbin, China

Minor **Harbin Institute of Technology**, Big Data Management and Applications — Harbin, China

Publications

Feature-Space Planes Searcher: A Universal Domain Adaptation Framework for Interpretability and Computational Efficiency — June 2026

Published in IEEE TPAMI. Feature-space unsupervised domain adaptation with frozen encoders, decision-boundary adaptation, and efficient full-dataset optimization.

Zhitong Cheng, Yiran Jiang, Yulong Ge, Yufeng Li, Zhongheng Qin, Rongzhi Lin, Jianwei Ma

10.1109/TPAMI.2026.3703974 (IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), Early Access)

Technical Skills

Mathematics and Theory: Computational mathematics, applied mathematics, linear algebra, probability, statistical learning theory, variational inference

Large Language Models: Foundations of LLM fine-tuning and practical LoRA fine-tuning

Generative Modeling: Diffusion models for natural-image generation

Embedded Systems: STM32 development and SolidWorks mechanical modeling

Languages: Chinese (native); English (CET-6 579, CET-4 593)

Selected Honors

- People's Scholarship; Outstanding Youth League Member; Outstanding Student; Special Graduate Academic Scholarship

Projects

Feature-Space Planes Searcher (FPS) for Unsupervised Domain Adaptation

Feature-space unsupervised domain adaptation framework that keeps pretrained encoders frozen and adapts decision boundaries using reusable H5 feature banks.

- Public PyTorch implementation for efficient and interpretable feature-space domain adaptation.
- Related work published in IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), Early Access, 2026.

Martian Geological Fault Mapping

Contributed to a collaborative project applying foundation models to Martian geological fault mapping.

- Assisted with vectorization and standardization of geological fault data.
- Contributed to selected parts of the fault-mapping workflow.

Efficient Sampling Algorithms for Diffusion Generative Models

Undergraduate thesis on improving the sampling efficiency of diffusion models.

- Investigated and designed algorithms targeting a central efficiency challenge in diffusion-model sampling.

Generative-Model-Based Ultrasound Medical Image Translation

Cross-modal medical-image translation research using generative models.

- Explored generation of diagnostically informative images to address limitations of single-modality ultrasound data.

Open Source Projects

AI Feed Tracker

AI subscription service that summarizes time-sensitive updates close to real time.

- Developed the service and its push-based information workflow.

Open-Source Contributions

Contributor to community projects including OpenChamber.

Volunteer Experience

- **Computer 110 Club Lead (2020-2022):** Supported on-campus computer diagnostics and wrote new-semester computer-purchasing guides for students.

Personal Interests

- Photography; fitness; basketball; badminton; table tennis; exploring and tinkering with technology products